

Scalable Reputation-Weighted Consensus for Byzantine-Robust Decentralized LLM Oracles

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Abstract—Deploying Large Language Models (LLMs) as multi-agent systems mitigates the hallucinations and single-point-of-failure risks of individual models. However, existing leader-based architectures remain vulnerable to malicious leaders. While leaderless architectures resolve this, they introduce an evaluator reliability problem, where egalitarian voting allows high-variance or Byzantine actors to degrade consensus, and an unscalable $O(N^2)$ communication bottleneck. We propose a scalable, meritocratic framework that hardens leaderless architectures against these threats. First, we introduce Softmax-based sharding mechanism to probabilistically select evaluator committees based on reputation, reducing communication complexity to a constant $O(K^2)$. Second, we adapt the Exponential Moving Average (EMA) to dynamically track historical agent performance, integrating reputation-based voting weights into a Weighted Geometric Median (WGM) consensus algorithm. Empirical evaluations demonstrate our framework reduces mean execution latency by 21.0% and improves consensus accuracy by +2.0 percentage points. Furthermore, the system proactively suppresses malicious participation to 26.6% during 40% network infiltrations and dynamically constructs a Meritocratic Safety Margin to survive extreme 50% coordinated Byzantine attacks. Ultimately, this approach provides a highly robust, low-latency data feed for blockchain oracles.

Index Terms—decentralized oracle, LLM, leaderless multi-agent system, reputation-weighted consensus, byzantine robustness, scalable

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated remarkable capabilities across a diverse range of complex tasks. However, their deployment as single-agent systems is constrained by inherent limitations, such as a propensity for hallucination and the vulnerability of a single point of failure [1]. Consequently, multi-agent LLM systems, wherein multiple agents collaborate to solve problems, have emerged as a promising paradigm. By leveraging mechanisms like quorum-based voting and multi-round debates, these collaborative frameworks can overcome the weaknesses of individual models, leading to significant improvements in overall answer quality. The migration of these systems to open, decentralized environments such as public blockchain networks, however, introduces a critical set of challenges, particularly the need of Byzantine fault tolerance against malicious agents.

Existing Byzantine-robust solutions predominantly rely on leader-driven architectures, which introduce two fundamental vulnerabilities. First, these system are highly susceptible to targeted attack on the leader. If leaders are Byzantine, the system suffer repeated protocol restarts and dramatic latency increases or even a failure given the high intrinsic latency of LLM invocations [2], [3]. Second, quorum-based voting on a single leader’s proposal can result in sub-optimal answer selection, accepting underperforming responses even when superior alternatives exist elsewhere in the network.

As a foundational advance, the DecentLLMs protocol was recently proposed to directly address the flaws of leader-based systems [1]. It introduces a novel, leaderless architecture that decouples the roles of answer generation and evaluation. In this model, a group of “worker” agents generate potential answers in parallel, and a separate group of “evaluator” agents independently score the quality of each proposed answer. The final consensus answer is then selected by applying a Byzantine-robust aggregation technique specifically, the Geometric Median (GM), to the scores provided by the evaluators. This design successfully mitigates the latency and answer-quality issues associated with Byzantine leaders. Yet, in solving one problem, it has brought a more subtle challenge into focus.

The central problem motivating our research is a newly identified frontier exposed by the DecentLLMs protocol itself: the *evaluator reliability problem*. While the leaderless architecture effectively neutralizes the threat of a Byzantine leader, experimental results reveal a critical vulnerability in the evaluation phase. Findings show that even honest evaluator agents can fail to score answers reliably, exhibiting high variance in their assessments and, in some cases, assigning low scores to correct answers or high scores to incorrect ones [1]. This high variance among honest agents reduces the effectiveness of standard Geometric Median aggregation, treating the input of a consistently accurate expert the same as that of an unreliable novice. The original authors conclude that improving the “reliability of the evaluators scoring mechanism” is an vital direction for future work. Moreover, leaderless systems require every evaluator to broadcast their scores to every other evaluator to calculate the consensus.

This creates an $O(N^2)$ communication nightmare, meaning the system will become terribly slow and expensive as the network grows.

Our Approach. To overcome these limitations, we propose a scalable, meritocratic framework that hardens leaderless architectures against both high-variance and Byzantine agents. First, to resolve the $O(N^2)$ communication bottleneck, we introduce a Softmax-based sharding mechanism. By probabilistically selecting a fixed-size committee (K) of evaluators based on their on-chain reputation score, we reduce the communication complexity to a constant $O(K^2)$ while ensuring fair network participations. Second, to solve the evaluator reliability problem, we integrate an Exponential Moving Average (EMA) trust model into a novel Weighted Geometric Median (WGM) consensus algorithm. By mathematically scaling each evaluator’s voting influence in proportion to their historical accuracy, our framework naturally filters out anomalies and malicious attacks.

TABLE I
COMPARISON OF DECENTRALIZED LLM MULTI-AGENT FRAMEWORKS

Dimension	BlockAgents [2]	MultiLLM [3]	DecentLLMs [1]	Ours
Architecture	Leader	Leader	Leaderless	Leaderless
Weighted Voting	✗	✓	✗	✓
Byzantine-Robust	✓	✓	✓	✓
Sharded Selection	✗	▲	✗	✓
Meritocratic Consensus	✗	▲	✗	✓
Communication	PoT	WBFT	GM	WGM
Validation	$O(N^2)$	$O(K)$	$O(N^2)$	$O(K^2)$
	Benchmark	Simulation	Benchmark	Both

Symbols: ✓ = Full support; ▲ = Partial; ✗ = Not supported.
Abbreviation: PoT = Proof-of-Thought; WBFT = Weighted BFT; GM = Geometric Median; WGM = Weighted GM.;

Contributions. Our primary contributions are summarized as follows:

- We proposed a consensus algorithm called Weighted Geometric Median (WGM) that makes agents influence to the network is mathematically proportional to their historical reliability.
- We establish a method to dynamically track and update agent performance over time. This allows our system to assign precise voting weights based on historical accuracy rather than single-round performance.
- We use sharding mechanism for choosing evaluator to reduce communication bottleneck.
- We build an open source prototype & provide an simulation to evaluate the performance of proposed framework.

The rest of this paper is organized as follows. Section II outlines relevant research on LLM-based multi-agent coordination and Byzantine-robust distributed systems. Section III introduces the foundational system architecture, defines problem, and formalizes the Byzantine threat model. Section IV details our proposed framework, explain the Softmax Sharding

Engine, the Reputation-Weighted Geometric Median (WGM) aggregation, and the Exponential Moving Average (EMA) update rule. Section V presents our empirical evaluation, analyzing the system’s execution latency reduction, absolute fault tolerance thresholds with the Meritocratic Safety Margin, and dynamic threat mitigation capabilities. Section VI discusses the broader architecture implications of our approach, acknowledge prototype limitations, and proposes future directions. Finally, Section VII concludes the research.

II. RELATED WORK

Recent advancements in decentralized Large Language Model (LLM) networks have focused on mitigating the risks of hallucinations and malicious behaviors through multi-agent collaboration. However, existing frameworks face critical trade-offs between latency, evaluator reliability, and scalability. We categorize the related literature into three primary domains to highlight the specific gap our research addresses.

A. Leader-Driven Oracle Protocols

To tolerate Byzantine (malicious) agents, early decentralized multi-agent systems adopted leader-driven consensus architectures. For instance, the BlockAgents [2] introduces a Proof-of-Thought (PoT) consensus mechanism combined with stake-based miner designation, where a designated leader proposes a block of answers that undergoes a multi-round, debate-style voting process. Similarly, the Trusted MultiLLMN [3] employs a Weighted Byzantine Fault Tolerance (WBFT) consensus, utilizing a HotStuff-like two-phase prepare/commit process with a rotating leader.

While these protocols successfully prevent basic poisoning attacks, their reliance on a central leader introduces severe operational vulnerabilities. If the designated leader is Byzantine or highly incompetent, the system repeatedly fails to achieve the required quorum (e.g., a two-thirds majority), forcing the protocol to restart or rotate leaders [1]. Given the high intrinsic latency of LLM invocations, these consecutive leader failures result in unacceptable consensus latency. Furthermore, because these system vote sequentially on a single leader’s proposal, they are prone to finalizing an underperforming answer simply because it reached a quorum, even when superior alternatives exist in the network.

B. Leaderless Coordination

To resolve the latency bottlenecks and single points of failure in leader-based models, recent research has pivoted toward leaderless architectures. The DecentLLMs protocol solve leader-based model problem by completely decoupling the answer generation and evaluation phases [1]. In this framework, worker agents generate answer in parallel while a separate group of evaluator agents independently scores all worker output. The final answer is then determined by applying the Geometric Median (GM) algorithm to evaluators multidimensional score vectors.

Although DecentLLMs effectively neutralizes targeted attacks on leaders and evaluates multiple answers simultaneously, it introduces two problems. First, it suffers from the

“evaluator reliability problem.” The standard GM algorithm operates on an egalitarian, treating all evaluators equally. Experimental evaluations of DecentLLMs reveal that even honest evaluators exhibit high variance which frequently assigning low scores to correct answers [1]. Treating an unreliable node equally to reputable node degrades the accuracy of the final consensus. Second, the protocol require every evaluator to broadcast their score vectors to all other evaluators [1], resulting in a quadratic $O(N^2)$ communication overhead that limits the network scalability.

C. Reputation and Trust Mechanisms

To address agent unreliability, trust and reputation mechanism have been proposed to penalize malicious behaviour and incentivize honest participation. [4] proposed a blockchain-driven, behaviour-shaping incentive mechanism for decentralized LLM multi-agent systems. Their framework dynamically updates an agent’s reputation score utilizing an Exponential Moving Average (EMA) rule based on historical task performance.

While this mechanism successfully utilizes reputation to calculate matching scores for initial task allocation, its application remains largely confined to a decentralized task marketplace model [4]. Current literature lacks a methodology to integrate dynamic, EMA-based trust metrics directly into the spatial aggregation logic of a leaderless consensus mechanism (like the Geometric Median).

Summary. Existing systems force a compromise: leader-driven models offer structured security but suffer from latency and bottlenecks, while leaderless models are fast but mathematically vulnerable to unscalable communication and high-variance evaluators. Our research directly bridges this gap by integrating a dynamic reputation mechanism into a leaderless architecture, using reputation weights to solve the evaluator reliability problem and Softmax-based sharding to solve the scalability bottleneck.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network and Agent Model

Our proposed framework operates within a decentralized, peer-to-peer network that strictly enforces a leaderless architecture. By eliminating the reliance on a designated leader or coordinator agent, the system mitigates the single-point-of-failure vulnerabilities and the severe consensus latency spikes cause by consecutive targeted Byzantine Attacks, which frequently degrades state-of-the-art leader-driven protocol.

Within this decentralized environment, we define a total pool of N large language model (LLM) based agents. For any given task, these agents operate under two distinct, parallel functional roles:

- **Worker Agents (W):** A group of agents denoted as $W = \{w_1, w_2, \dots, w_{N_w}\}$ is responsible for receiving the user’s prompt and generating proposed answer. Crucially, worker agents independently generate their responses in parallel. This architecture guarantees horizontal scalability, allowing the network to increase throughput simply

by allocating additional GPU resources to generate more answer simultaneously.

- **Evaluator Agents (E):** A selected “Jury” committee of agents denoted as $E = \{e_1, e_2, \dots, e_{N_e}\}$ is tasked with independently assessing and quantitatively scoring the response generated by the workers.

To guarantee security and transparency, the multi-agent network is secured by a Blockchain Layer. This layer acts as an immutable, globally auditable registry that records agent registration and round logs. Most importantly for our framework, the blockchain persistently stores and updates the dynamic Reputation Score (T) for every evaluator agent. By securely saved these reputation scores on-chain, the protocol ensures that our consensus remains entirely transparent and tamper-proof.

B. The Evaluation Space

In our proposed framework, evaluating a worker’s answer is not a simple binary vote, but rather a multidimensional quantitative assessment. To capture the complex nuances of natural language generation, we mathematically define the evaluation space across $C = 5$ distinct, continuous criteria.

When a selected evaluator agent assesses a worker’s proposed answer, it calculates a specific score for each of the following five dimensions [5]:

- 1) **Factual Contradiction:** The degree to which the answer contradicts established facts.
- 2) **Factual Fabrication:** The extent to which the answer invents or hallucinates non-existent information.
- 3) **Instruction Inconsistency:** How much the answer deviates from the user’s original prompt.
- 4) **Context Inconsistency:** The degree to which the answer strays from the original context of the question
- 5) **Logical Inconsistency:** The presence of incorrect logical argumentation or flawed reasoning steps.

According to the [1], each of these $C = 5$ criteria is strictly evaluated on a continuous scale from 0 to 20. On this scale, a score approaching 20 indicates high quality, a score approaching 0 indicates poor quality, and a score of 10 represents neutrality.

The evaluation produced by an evaluator agent e_i for an answer generated by a worker agent w_j is represented as an 5-dimensional score vector, denoted as $A_{ij} \in \mathbb{R}^5$. Because every evaluation is partial coordinate in this 5D space rather than a single scalar value, our framework must rely on the L_2 norm (Euclidean distance) to accurately measure the geometric difference between these vectors during the consensus phase.

C. The Threat Model

Because our proposed decentralized oracle is designed to operate on open, permissionless blockchain networks, we must assume a highly adversarial environment. In such peer-to-peer networks, any participant can join the system, meaning we cannot guarantee that all agents will behave honestly or rationally. We classify any agent that intentionally deviates from the protocol as a Byzantine Adversary. A Byzantine agent

can behave completely arbitrarily to disrupt the consensus, degrade the quality of the final output, or manipulate the system for its own benefit.

Within our leaderless architecture, these adversaries can launch sophisticated attacks across both functional roles:

- **Byzantine Worker Agents:** Instead of attempting to answer the user’s prompt accurately, a malicious worker might inject completely irrelevant advertisements, non-sensical data, or execute backdoor attacks by inserting hidden trigger phrases into their proposed text.
- **Byzantine Evaluator Agents:** Malicious evaluator pose a severe threat because they can actively manipulate the 5-dimensional evaluation phase to subvert the oracle’s output. Specifically, they can execute coordinated collusion attacks. In this scenario, Byzantine evaluators deliberately assign perfect scores (e.g., a 20 across all criteria) to the poisoned, garbage answers generated by Byzantine workers, while simultaneously artificially suppressing honest workers by assigning them score of 0. Furthermore, adversaries may utilize oscillation attacks, where they intermittently behave honestly to farm reputation before suddenly leveraging that accumulated weight to execute a targeted attack.

To guarantee the security of the oracle against these coordinated attacks, our framework mathematically bounds the maximum number of adversaries the system can withstand. Because our consensus engine adapts the Geometric Median algorithm to aggregate the multidimensional score vectors, the protocol ensures absolute Byzantine resilience provided there is a strict honest majority within the selected ”Jury” committee. Formally, if our shading engine selects a committee of size K , the system is mathematically proven to tolerate up to $f \leq \lfloor \frac{n-1}{2} \rfloor$ Byzantine evaluators. However, our proposed framework fundamentally alters this constraint. Because the consensus engine utilizes the Reputation-Weighted Geometric Median, the system’s fault tolerance is no longer dictated by the raw number of malicious agents, but by their accumulated spatial influence. Therefore, our system is mathematically proven to maintain secure spatial consensus as long as the Honest Weight Fraction (Λ) strictly satisfies the majority condition:

$$\Lambda = \frac{\sum w_{honest}}{\sum w_{total}} \geq 0.5$$

This dynamic threshold allows the oracle to tolerate advanced coordinated attacks by systematically stripping the voting weights of Byzantine actors via the Exponential Moving Average (EMA) update rule, ensuring Λ remains safely above the breakdown point.

D. The Problem Formulation

While adopting a leaderless architecture successfully neutralized the threat of consecutive targeted attacks against a single coordinator agent, deploying such a system at scale

exposes two critical vulnerabilities inherent in current state-of-the-art baseline protocols. We formally define these limitations as the Evaluator Reliability Problem and the Scalability Bottleneck, which our research aims to solve.

1) The Evaluator Reliability Problem

Standard leaderless protocols, such as our primary baseline DecentLLMs [1], utilize an egalitarian spatial aggregation method (standard Geometric Median) to calculate consensus. The fundamental flaw in this approach is that it mathematically assumes all non-Byzantine (honest) vectors are equally reliable and grants them equal weight in the spatial calculation.

However, due to the inherent stochastic nature of LLM, even completely honest evaluator agents naturally exhibit high variance in their accuracy. During the evaluation phase, an honest but inconsistent agent might incorrectly assign a low score to a perfectly accurate answer, or mistakenly assign a high score to a hallucinated answer. By treating all agents equally, the standard Geometric Median fails to distinguish between a consistent & inconsistent evaluator. The problem we must formulate and solve is how to mathematically transform this egalitarian voting model into a dynamic, meritocratic system where an evaluator’s spatial influence over the consensus center is proportional to their historical accuracy.

2) The Scalability Bottleneck

The second problem lies in the communication complexity required to reach a leaderless consensus. In a flat, unsharded leaderless protocol, every evaluator agent in the network must independently score the answers generated by every worker agent in for formulate their vectors. If the total agent pool consists of N agents, this ”all-to-all” evaluation creates a communication complexity of $O(N^2)$.

As the decentralized network grows to accommodate more agents, this exponentially exploding communication overhead results in latency spikes. Therefore, the second problem we must solve is how to mathematically decouple the total network size (N) from the consensus requirements, replacing the $O(N^2)$ bottleneck with a flat, manageable communication cost without permanently excluding new agents from participating.

IV. PROPOSED FRAMEWORK

A. Architecture Overview

The proposed framework introduces a decentralized, leaderless multi-agent architecture designed to provide highly accurate and Byzantine-robust data feeds. To overcome the latency and single-point-of-failure vulnerabilities in traditional leader-driven oracle protocols, our system entirely decouples the roles of task execution and quality evaluation. As illustrated in Figure 1, the end-to-end operational workflow is divided into three interconnected layers: Off-Chain Agent Layer, Consensus Layer, and On-Chain Blockchain Layer.

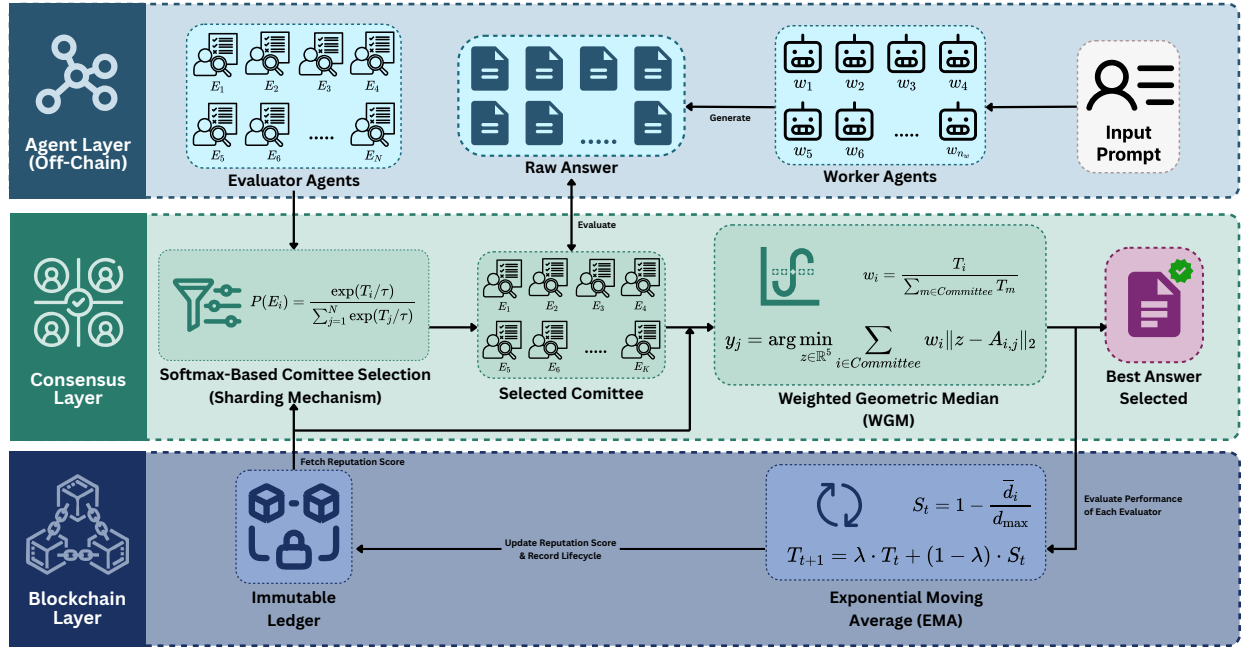


Fig. 1. Architecture overview

The lifecycle of a single consensus round proceeds through the following sequential phases:

1) **Task Initialization and Parallel Generation**

The process is initiated when a user submits a query or prompt to the decentralized network. Within the Agent Layer, a group of dedicated LLM instances, designated as *Worker Agents* (W), receive this prompt. To maximize throughput and mitigate the hallucination risks of any single model, these workers process the prompt and generate their respective answers concurrently.

2) **Softmax-Based Committee Sharding**

To resolve the $O(N^2)$ communication scalability bottleneck of standard leaderless protocols, the system does not require all available evaluators to score the worker's answer. Instead, the Consensus Layer triggers the *Sharding Engine*. This module fetches the historical reputation scores (T_i) of all available evaluator agents from the immutable blockchain registry. Using a Softmax probability distribution, the engine dynamically selects a fixed-size "Jury" or committee of K evaluators. This ensures that network communication remains constant at $O(K^2)$ while probabilistically favoring historically reliable agents.

3) **Committee Evaluation and Scoring**

Once the committee is established, the selected K *Evaluator Agents* (E) independently assess the quality of the N_w answers generated by the workers. As defined in the section III, evaluator rigorously grades every answer across the $C = 5$ distinct criteria & resulting a 5-dimensional evaluation score vector. These individual 5D vectors from the K committee members are then compiled into a set of multidimensional score matrices

(one matrix for each worker's answer), which are subsequently passed to the Consensus Engine for spatial aggregation.

4) **Reputation-Weighted Consensus and Decision-Making**

Rather than treating all evaluations equally, our framework employs a meritocratic aggregation mechanism to calculate the finalized answer. Before aggregation, each evaluator's contribution to the score matrix is mathematically scaled by a normalized weight (w_i), which is directly proportional to their on-chain reputation. The *Consensus Engine* then applies a Weighted Geometric Median (WGM) algorithm to this matrix. This spatial optimization finds the most robust central consensus score, effectively neutralizing high-variance anomalies or coordinated Byzantine attacks. The worker's answer corresponding to the highest aggregated robust score is selected as the finalized, accurate answer and returned to the user.

5) **Post-Round Settlement and Trust Update**

Following the decision-making phase, the system evaluates the performance of the committee members. By calculating the spatial error (the difference between an individual evaluator's submitted score vector and the finalized WGM consensus score) the system derives a current round performance metric (S_t). This metric is fed into an Exponential Moving Average (EMA) update rule to recalculate each evaluator's reputation. These updated scores are subsequently recorded on the Blockchain Layer, ensuring that consistently accurate evaluators gain more influence in future rounds while unreliable actors lose voting weight).

B. Softmax-Based Committee Selection (Scalability)

A fundamental limitation of existing leaderless multi-agent architectures, such as the DecentLLMs [1], is the communication bottleneck caused by the all-to-all broadcast requirement. Because every evaluator must share its score vector with all other evaluators to calculate geometric median locally, the communication complexity scales quadratically at $O(N^2)$. To resolve this scalability bottleneck, our framework introduces a dynamic sharding mechanism that selects a fixed-size committee of K evaluators for each task. It will reduce the communication complexity to a constant $O(K^2)$.

However, selecting this committee presents a critical design trade-off between network security and fairness. Naive deterministic approach would create a closed system where new agents joining the network with a default initial reputation (e.g., $T = 0.5$) would never be selected over established agents (e.g., $T = 0.9$). It will make the new agents no opportunity to grow their reputation.

To solve this "cold start" problem, we adopt a probabilistic selection mechanism by adapting the task assignment probability distribution proposed in [4]. Instead of deterministic sorting, we apply a Softmax function to the reputation scores of all available evaluators. The probability $P(e_i)$ of selecting evaluator E_i for the committee is computed as:

$$P(e_i) = \frac{\exp(T_i/\tau)}{\sum_{j=1}^N \exp(T_j/\tau)}$$

where T_i denotes the current historical reputation score of evaluator i , N represents the total pool of available evaluators agents, and τ represents the temperature hyperparameter. Using these calculated probabilities, the system performs a weighted random sampling without replacement to select exactly K distinct evaluators.

This Softmax-based formulation mathematically balances security and exploration. The temperature parameter τ serves as a tunable threshold for:

- 1) Lower τ will amplify the differences in reputation score, making the selection process only selected highly reputable agents. It will maximize Byzantine robustness but limits diversity.
- 2) Higher τ will smooths the probability distribution, making the selection more uniform and allowing for broader exploration of the network.

Because the Softmax function ensures that lower-reputation agents maintain a mathematically non-zero probability of selection, it will provide a fair environment where new or recovering agents are occasionally drafted into the committee.

C. Reputation-Weighted Geometric Median (Accuracy)

Following the Sharding phase, the system isolates a selected committee of size K . Unlike the baseline DecentLLMs [1] which aggregates a single scalar per answer, our system requires each evaluator $i \in \text{Committee}$ to assess every worker's answer j across $C = 5$ distinct criteria established in

section III. This produces a multidimensional evaluation score vector $A_{ij} \in \mathbb{R}^5$ for each worker's answer.

While the baseline protocol relies on a standard Geometric Median (GM) to resist Byzantine influence, treating all evaluators equally leaves the consensus vulnerable to the evaluator reliability problem, where honest but high-variance agents drag down the accuracy of the final consensus. To overcome this, we incorporate the historical reputation weights (w_i) into the process. For each worker's answer j , the final 5 dimensional consensus score vector y_j is determined by finding the spatial point that minimizes the weighted sum of Euclidean distances to the K evaluators score vectors:

$$y_j = \arg \min_{z \in \mathbb{R}^5} \sum_{i \in \text{Committee}} w_i \|z - A_{ij}\|_2$$

$$w_i = \frac{T_i}{\sum_{m \in \text{Committee}} T_m}$$

where T_i represents the historical reputation score of evaluator i . This normalization ensures that historically reliable agents exert a mathematically stronger gravitational pull on the final consensus point in the 5D space.

Because this objective function lacks a general closed-form algebraic solution, we compute the Weighted Geometric Median (WGM) using an adapted, weighted version of Weiszfeld's algorithm [6], an iteratively reweighted least squares method. Let $y_{j,k}$ be the consensus estimate for answer j at iteration k . The algorithm proceeds through the following structured steps:

- 1) **Initialization:** To accelerate algorithmic convergence, the initial estimate $y_{j,0}$ is established by calculating the standard median of the source matrix containing the K vectors.
- 2) **Iterative Update:** The consensus estimate is iteratively refined using our modified reputation-weighted update rule:

$$y_{j,k+1} = \frac{\sum_{i=1}^K \frac{w_i \cdot A_{i,j}}{\|A_{i,j} - y_{j,k}\|_2}}{\sum_{i=1}^K \frac{w_i}{\|A_{i,j} - y_{j,k}\|_2}}$$

This calculation scales both numerator and denominator by the evaluator's specific reputation weight w_i , alongside the inverse of their Euclidean distance to the current estimate.

- 3) **Singularity Handling:** A known limitation of Weiszfeld's algorithm is that it becomes undefined if the current estimate $y_{j,k}$ lands precisely on an existing data point A_i , causing division by zero. We handle this zero-distance edge case by continuously checking if $\|A_{i,j} - y_{j,k}\|_2 < \epsilon$. If this condition is met, the algorithm snaps the current estimate directly to $A_{i,j}$ and successfully breaks the iteration loop.
- 4) **Convergence:** The iterations continue until the Euclidean distance between consecutive estimates falls below a script convergence tolerance threshold ϵ (set to 10^{-5} or until the process reaches a predefined safety limit of 1000 iterations).

Upon successful convergence, the algorithm return a finalized 5D consensus score y_j vector for the worker’s answer. To determine the winning proposal, the Consensus Engine sums the 5 dimensions of y_j to produce a single, scalar robust score bounded between 0 and 100. The system evaluates the robust score across all N_w workers and selects the answer with the highest total scores as the optimal, verified output.

In the event that two or more workers submit answers that receive the exact same highest robust score, the system employs a deterministic cryptographic tie-breaker. The protocol concatenates the raw text of each tied answer with the hash of the most recent blockchain block, hashes this combined string, and automatically selects the answer that generates the largest resulting hash value.

By mathematically coupling Weiszfeld’s 5-dimensional spatial optimization with reputation weight make this algorithm effectively filters out compromising behavior. If a Byzantine or high-variance evaluator attempt to attack by submitting extreme score vector, their geometric distance from the true median is neutralized by their reputation weight w_i .

D. Dynamic Reputation Update Mechanism

To maintain long-term agent accountability and enforce our meritocratic architecture, we implement a dynamic reputation update mechanism adapted from [4]. While the Softmax Sharding and Weighted Geometric Median (WGM) algorithms neutralize Byzantine threats spatially within a single round, the system must also regulate agent behavior temporally to prevent malicious nodes from continuously participating in future consensus committees.

After the Consensus Engine calculates the finalized spatial center y using the WGM, the system retroactively evaluates the performance of each evaluator agent who participated in the ”Jury” committee. In our framework, an evaluator’s spatial error is quantified by the average Euclidean distance ($\bar{d}_i$) between their proposed 5-dimensional evaluation vectors ($A_{i,j}$) and the finalized consensus centers (y_j) for all answers evaluated in that round.

To map this spatial error into a valid performance metric, the system applies an inverse min-max normalization strategy [7]. This maps the minimum possible distance to a maximum performance score of 1, and the maximum possible distance (d_{max}) to a score of 0:

$$S_t = 1 - \frac{\bar{d}_i}{d_{max}}$$

In this equation, d_{max} represents the maximum possible absolute distance within the 5D evaluation metric space. Because each of the $C = 5$ criteria operates on a strict scale from 0 to 20, the worst-case Euclidean distance is $d_{max} = \sqrt{5} \times 20^2 \approx 44.72$.

Once the performance score is calculated, the agent’s permanent Reputation Score (T) updated on the blockchain layer using an Exponential Moving Average (EMA) rule:

$$T_{t+1} = \lambda \cdot T_t + (1 - \lambda) \cdot S_t$$

In this equation, $\lambda \in [0, 1)$ serves as a smoothing coefficient that dictates the historical inertia of the reputation system. A higher λ value provides more stability by relying heavily on past reputation, where a lower λ makes the score highly sensitive and responsive to recent performance.

This EMA formulation is critical for two reasons. First, it guarantees that reputation scores remain strictly bounded within the interval, preserving mathematical consistency for the subsequent round’s Softmax selection. Second, it introduces a natural ”forgetting mechanism”. While this allows honest agents to eventually recover from occasional, unintentional hallucinations, it acts as a severe penalty for Byzantine agents. By systematically tracking extreme variance, the EMA rule mathematically decrease the T scores of attackers. Because an agent’s T score directly dictates both their voting weight in the WGM and their probability of being selected by the Sharding Engine, this dynamic update mechanism permanently strips malicious agents of their influence.

V. EXPERIMENT EVALUATION

A. Experimental Setup

Implementation Environment and Reproducibility. To evaluate the proposed framework against the baseline (DecentLLM), we implemented an asynchronous pipeline in Python 3.12.13 that models a decentralized multi-agent network over the OpenRouter API. To execute this simulations, the experimental prototype was deployed on a lightweight Virtual Private Server (VPS) running the Ubuntu 22.04, configured with 2 vCPUs, 4GB of RAM, and 60 GB of disk storage. The multi-agent interactions are handled via LangChain’s ChatOpenAI client [8], which coordinates the generation and evaluation workloads concurrently. For full transparency, reproducibility, and detailed information regarding the implementation mechanics, the complete source code of our prototype is publicly available in our GitHub repository here.

The Agent Pool. For this prototype, we deployed a heterogeneous pool of 30 different LLM across 9 different providers, dividing them strictly into 15 worker agents responsible for generating candidate answers and 15 evaluator agents responsible for scoring them. A complete enumeration of the specific model that we used is detailed in A

The Dataset & Parameters. All experiments were evaluated over 50 consecutive rounds. To rigorously test the framework ability to provide accurate data, we utilized a hybrid dataset comprising 50 questions evenly split between two benchmark: TruthfulQA [9], which tests factual accuracy and susceptibility to common misconceptions, and ARC-Easy [10], which evaluates scientific reasoning. The dataset questions were shuffled per-run using a seeded random number generator to ensure reproducibility across trials. For the params configuration, we initialized all agents with a default reputation score of 0.5, set the Softmax temperature τ to 0.3, set the sharded ”Jury” committee size to $K = 10$, and utilized a smoothing factor of $\lambda = 0.3$ for the EMA reputation update rule.

On-Chain Reputation Layer and Smart Contract Prototype. To provide a tamper-evident audit trail and deterministic conflict resolution for the multi-agent network, the experimental prototype integrates a blockchain layer utilizing a Solidity smart contract deployed on a local Ethereum-compatible Anvil testnet. This on-chain prototypes serves three critical functions within our architecture:

- 1) **Reputation Anchoring and Round Provenance:** Agent reputation scores (T) are mapped to integer values and anchored directly on-chain. Upon the completion of each consensus round, the orchestrator submits a batch transaction to atomically update the participating agents reputation and log the round’s metadata. This creates an immutable, cryptographically secure audit trail.
- 2) **Deterministic On-Chain Tie-Breaking:** In condition where multiple worker agents receive identical robust scores from the Weighted Geometric Median, the consensus engine resolves the tie deterministically using a cryptographic hash function. The system merge its answer with blockhash, hash it and selects the answer producing the largest hash value as integer. This logic is natively replicated within the smart contract’s verification functions, ensuring that the tie-breaker mechanism can be independently verified on-chain without trusting a centralized random number generator.

The Baselines. We conducted head-to-head evaluations between two primary system configurations:

- **DecentLLM Baseline [1]:** This configuration models the current state-of-the-art. Sharding is disabled, meaning all N evaluators are forced to participate in the consensus calculation. It utilizes an egalitarian weight mode where every evaluator receives an identical voting weight.
- **Our Proposed System:** Sharding is enabled. The system probabilistically samples a smaller committee of K evaluators utilizing Softmax over their reputation scores. It employs a meritocratic weight mode where an evaluator’s spatial voting influence is proportional to their normalized reputation, updated continuously via the EMA rule.

Passive Shadow Tracking for Latent Safety Margin Analysis. To enable a direct mathematical comparison of internal safety margins across fundamentally different architectures, we implemented a passive shadow tracking methodology during the baseline evaluations. During the DecentLLMs experiments, the baseline was executed in its native, strictly stateless egalitarian configuration, where all N nodes maintained a rigid voting weight of $w = \frac{1}{N}$. However, we allowed our proposed Exponential Moving Average (EMA) engine to runs as an isolated background observer. This passive logging did not influence the baseline’s spatial aggregation, evaluator selection, or final consensus outputs. Instead, it served purely as an independent diagnostic metric to extract the baseline’s latent Honest Weight Fraction (Λ). By running this shadow metric, we were able to empirically quantify the hidden structural degradation of baseline’s safety margin

as the Byzantine threat scaled, without violating its stateless operational mechanics.

The Threats Models. To empirically validate the Byzantine fault tolerance thresholds of the proposed framework, we programmed four specific adversarial evaluator behaviors to inject into the network:

- 1) **Random Attack:** Evaluators inject pure noise by replacing every score dimensions with a uniformly random value between 0 and 20.
- 2) **Inverse Attack:** Evaluators systematically invert the quality score, rating good answers poorly and bad answers highly by assigning a score of $20.0 - \text{truescore}$.
- 3) **Oscillation Attack:** Evaluators attempt to evade detection by alternating between an honest phase and a random attack phase based on a configurable round period.
- 4) **Amplification Collusion:** The most extreme coordinated threat, where colluding evaluators automatically assign perfect score of 20 across all dimensions to malicious workers while aggressively suppressing honest workers with scores of 0. In the adversarial experiments, the malicious evaluators were selected randomly form the pool based on target presence fractions to simulate varying severities of network compromise.

Simulation Constraints vs. Theoretical Deployment. In a real-world, theoretical deployment of the proposed decentralized oracle, the multi-agent system operates on an open, peer-to-peer blockchain network where each agent is hosted on physically separate nodes with dedicated GPU resources. In such a true leaderless architecture, worker and evaluator agents execute their LLM generation and scoring workloads in absolute parallel without inter-agent communication queues, thereby guaranteeing true horizontal scalability. However, for the scope of this experimental prototype, the decentralized multi-agent network was simulated locally and routed thorough a single centralized commercial API (OpenRouter). Because standard baseline protocols like DecentLLM required all N evaluator to score all N workers, simulating this ”all-to-all” communication triggered severe centralized rate limits (HTTP 429 errors) from the API provider. To practically simulate the multi-agent network without crashing the centralized provider, we implemented controlled asynchronous concurrency utilizing an `asyncio.Semaphore` with a capacity of 30. This mechanism physically restricted the simulation to a fixed number of in-flight API requests at any give millisecond, producing a natural queuing. It is critical to note that the absolute execution time recorded in these experiments include artificial queuing latency introduced by this semaphore. This constraint was applied equally across all experiment.

B. Scalability and Performance Analysis

To evaluate the computational overhead and communication efficiency of the proposed framework, we conducted a head-to-head performance analysis against the DecentLLMs [1] baseline in a clean, non-adversarial environment (0%

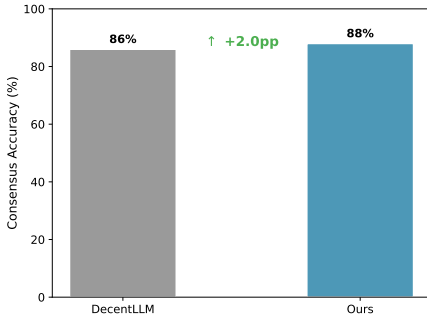


Fig. 2. Consensus accuracy comparison under no-attack conditions

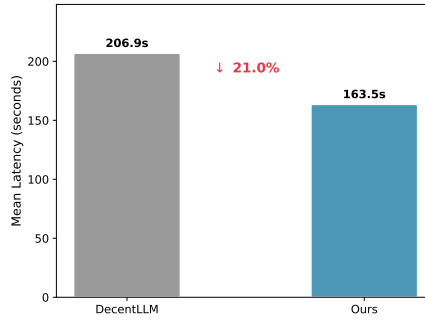


Fig. 3. Comparison of mean execution latency under no-attack conditions

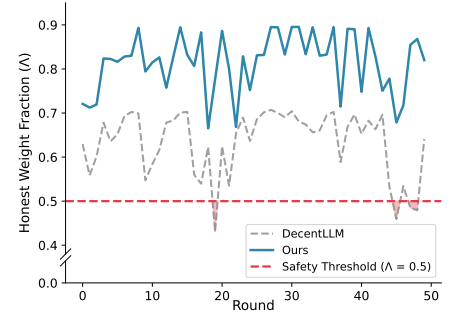


Fig. 4. Evolution of the honest weight fraction under a 50% Byzantine attack

malicious agents). By isolating the systems from Byzantine attacks, this evaluation specifically tests the effectiveness of the Softmax-Based Committee Selection engine in resolving the scalability bottlenecks inherent in flat leaderless protocols, while concurrently demonstrating the natural variance filtering of our Reputation-Weighted Consensus.

Overcoming the Communication Bottleneck. In standard leaderless architectures such as the DecentLLMs baseline [1], the consensus mechanism operates without sharding. This requires an “all-to-all” evaluation phase where every evaluator agent in the network pool (N) must independently score every generated answer, resulting in an unsustainable communication complexity of $O(N^2)$. As demonstrated in Figure 3, processing 50 consecutive rounds under this un-sharded egalitarian architecture resulted in a mean execution latency of 206.9 seconds per round.

On the other hand, our proposed framework utilizes Softmax-based sharding to dynamically isolate a smaller, fixed-size “Jury” committee of $K = 10$ evaluators based on their reputation scores. By mathematically constraining the Weighted Geometric Median calculation to only this selected committee, the system successfully flattens the communication complexity from $O(N^2)$ down to a manageable $O(K^2)$. The experiment result validate this theoretical advantage, our sharded architecture achieved a much faster mean execution latency of 163.5 seconds per round. This represents a highly significant 21.0% reduction in mean latency. Refer to the simulation constraints in V-A, while both systems were subjected to the exact same asynchronous API queuing limitations, this relative 21.0% reduction proves that our sharding engine optimize network throughput for real-world deployment.

Solving the Evaluator Reliability Problem. Crucially, this substantial reduction in execution latency was achieved without degrading the quality of the finalized output. In fact, our framework empirically outperformed the baseline’s accuracy. Over the TruthfulQA [9] and ARC-Easy [10] hybrid benchmark dataset, the un-sharded and equal weight DecentLLMs [1] baseline achieved an overall accuracy of 86%, where our proposed framework achieved 88% reflecting a +2.0 percentage point improvement.

This accuracy improvement perfectly validates our core

mathematical solution to the Evaluator Reliability Problem. Due to stochastic nature of LLM, even completely honest evaluator agents naturally exhibit high variance and occasional hallucinations. Because the DecentLLMs [1] baseline utilizes an egalitarian Geometric Median, it mathematically treats all non-Byzantine agents equally, granting these temporary, unintentional hallucinations equal spatial weight and subtly dragging the consensus off-target.

Our-framework dynamically tracked this natural variance using the EMA update rule. By applying normalized reputation score (T) to the spatial calculation within the sharded committee, our system successfully dampened the influence of highly inconsistent but honest agents. This meritocratic weighting pulled the Weighted Geometric Median (WGM) closer to the absolute truth, yielding high overall accuracy even in a completely peaceful environment.

C. Byzantine Resilience and Fault Tolerance Thresholds

To evaluate the absolute security limits of the proposed framework, we subjected the architecture to extreme Byzantine conditions. Traditional unweighted leaderless protocols, rely on rigid, count-based Byzantine Fault Tolerance (BFT) threshold, where consensus is only mathematically guaranteed if the absolute number of malicious vectors $f \leq \lfloor \frac{n-1}{2} \rfloor$.

As established in our Thread Model (III-C), because our framework replaces egalitarian voting with Reputation-Weighted Consensus mechanism, its survival is no longer strictly dictated by the raw headcount of attackers. Instead, the system must simply guarantee that the Honest Weight Fraction remains above the mathematical breakdown point ($\Lambda > 0.5$). If the $\Lambda \leq 0.5$, the Byzantine vectors gain the majority of the gravitational pull, hijacking the spatial center and breaking the oracle.

The Evaluator Reliability Problem and the Meritocratic Safety Margin. While maintaining $\Lambda > 0.5$ is the theoretical requirement for survival, our empirical data reveals that in LLM-based multi-agent networks, raw mathematical survival is a necessary but insufficient condition. Due to stochastic nature of LLMs, honest evaluator agents inherently exhibit high variance and occasionally hallucinate incorrect scores. In an egalitarian baseline system, any natural hallucination by

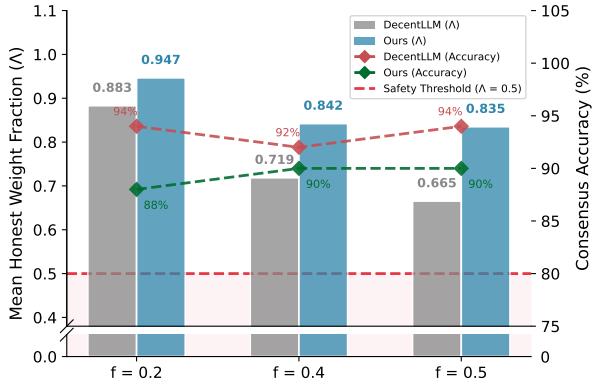


Fig. 5. Safety margin and accuracy under escalating random Byzantine attacks

an honest agent effectively subtracts voting weight from the "correct" side of the consensus. If a baseline system operates near the $\Lambda = 0.5$ limit, a minor cluster of honest hallucinations will instantly cause the effective Λ to drop below the threshold, giving the consensus to the Byzantine attackers.

To solve this evaluator reliability problem, our framework utilizes the Exponential Moving Average (EMA) update rule to proactively suppress the weights of highly variant agents, create a "Meritocratic Safety Margin" that pushes Λ safely away from the mathematical breaking point.

Dynamic Healing at the 50% Limit (Amplification Attack). To physically observe the creation of this safety margin, we evaluated both system under an Extreme 50% Amplification Attack. The evolution of the Honest Weight Fraction (Λ) over 50 consecutive consensus rounds is illustrated in Figure 4.

As demonstrated in Figure 4, the unweighted DecentLLMs baseline lacks an active tracking mechanism and therefore lacks a safety margin. Because the network was saturated with 50% malicious agents, the baseline's Λ violently fluctuated dangerously close to the 0.5 limit, fatally dipping below the threshold during 4 separate rounds. Consequently, the baseline overall accuracy is heavily degraded to 70%.

In contrast, our proposed framework exhibits profound dynamic healing. In Round 1, due to the "Cold Start" vulnerability where all agents initialize with a default $T = 0.5$, the system temporarily fitted with 0.5 boundary. However, in subsequent rounds, the EMA mechanism successfully detected the extreme Euclidean distance of Byzantine amplification vectors. By systematically reduce the reputation scores of the attackers, the system stripped their voting influence, aggressively driving the Honest Weight Fraction upward. Over the 50 rounds, our framework pushed the average honest weight to $\Lambda_c = 0.815$. By dynamically building this 0.315 Meritocratic Safety Margin, the proposed architecture successfully absorbed the targeted 50% Byzantine attack and the natural variance of honest LLMs. Ultimately, our framework maintained a highly resilient overall accuracy of 76% empirically outperforming the baseline's 70% under the exact same extreme adversarial conditions.

Scalability and the Illusion of Passive Survival. To further

investigate the correlation between the Mean Honest Weight Fraction (Λ_c) and overarching oracle reliability, we evaluated both system across a progression of Random Attacks, scaling the Byzantine threat from $f = 0.2$ up to the mathematical limit of $f = 0.5$. The resulting dual-axis chart in Figure 5 compares the Average Honest Weight Fraction (Λ_c) against the system's Overall Accuracy.

As illustrated by the data, the egalitarian baseline (DecentLLMs) maintained a high raw accuracy against uniformly distributed random noise, sustaining 94% accuracy even at $f = 0.5$ limit. However, the dual-axis visualization exposes this performance metric is highly deceptive. Because the baseline operates without a sharding mechanism, it achieves this accuracy by brute-forcing the consensus, forcing all N agents in the network to participate in an unsustainable $O(N^2)$ all-to-all communication overhead. Furthermore, while the baseline's accuracy line remains deceptively high, its passive Honest Weight Fraction bars exhibit a severe and steady collapse as the threat scales. The baseline's safety margin degrades from $\Lambda_c = 0.883$ at $f = 0.2$ down to $\Lambda_c = 0.665$ at $f = 0.5$, sinking dangerously close to the 0.5 Safety Threshold. As demonstrated previously in the Amplification Attack, relying on this decaying, passive margin leaves the baseline entirely defenseless the moment Byzantine vectors shift from random noise to targeted coordination.

In contrast, our proposed framework deliberately trades a negligible fraction of accuracy under simple random noise to guarantee scalable $O(K^2)$ execution and absolute Byzantine robustness. By utilizing the Softmax Sharding Engine, our framework restricts evaluations to a dynamic "Jury" committee. While relying on this smaller subset introduces a minor degree of natural LLM variance, resulting in a slight accuracy adjustment to 90% at $f = 0.5$, the system actively produce a safety buffer. Rather passively allowing Λ_c to decay, our Exponential Moving Average engine dynamically identified and suppressed the highly variant random attackers, proactively make a Honest Weight Fraction of $\Lambda_c = 0.835$.

Ultimately, this empirical progression proves that while brute-force egalitarian protocols can passively survive uncoordinated noise, they cannot safely scale. Our leaderless multi-agent system successfully hardened the network against both unintentional variance and coordinated attacks by permanently abandoning raw headcount in a favor of a mathematically defended Meritocratic Safety Margin.

D. Trustworthiness and Malicious Rate Reduction

In Section V-C, we empirically demonstrated that the proposed framework survives extreme Byzantine thresholds by mathematically defending its Honest Weight Fraction (Λ). To understand the exact mechanism driving this robust spatial defense, we must analyze the internal dynamics of the Consensus Layer. Specifically, we evaluate how the integration of Exponential Moving Average (EMA) update rule and the Softmax Sharding Engine proactively builds trustworthiness and physically reduces malicious influence over time.

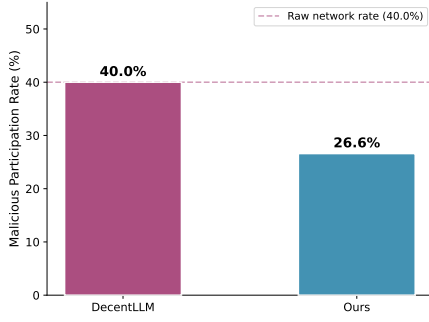


Fig. 6. Malicious committee participation rate for DecentLLM vs Ours under 40% Byzantine fraction

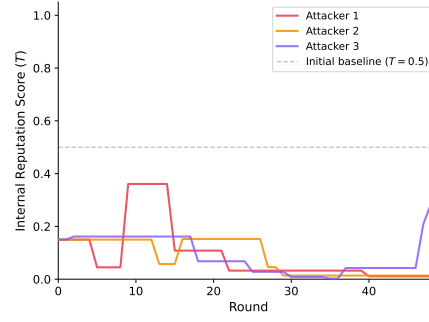


Fig. 7. Reputation trajectory (T) of malicious agents under the Inverse Attack.

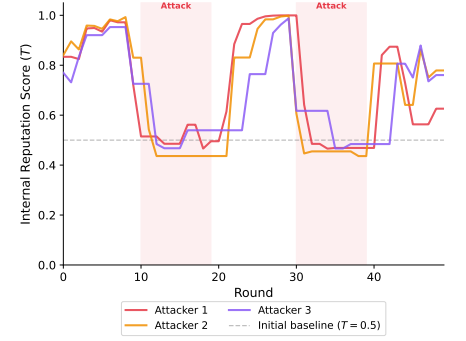


Fig. 8. Reputation trajectory (T) of malicious agents under the Oscillation Attack

Malicious Rate Reduction. In traditional egalitarian protocols, such as the DecentLLMs baseline, evaluators are selected uniformly, guaranteeing that Byzantine agents participate in the consensus at rate proportional to their raw network presence. While such baselines rely on recording raw transaction logs to the blockchain with the hope of manually reconfiguring the evaluator groups at some arbitrary point in the future, they possess no mechanism to restrict malicious participation in real-time.

Our framework solves this by coupling the internal Reputation Score (T) to physical participation probability. By utilizing the Softmax Sharding Engine, the probability of an evaluator being selected for the K -sized "Jury" committee is strictly proportional to their historical trustworthiness. To empirically validate this, we tracked the Malicious Committee Participation Rate during a 40% Byzantine threat scenario as illustrated in Figure 6.

As expected, because the baseline lacks a dynamic selection filter, malicious agents successfully infiltrated the evaluation committees at their raw network rate of 40%. In contrast, our proposed framework dynamically detected the anomalous Euclidean distances of the Byzantine vectors via the Weighted Geometric Median (WGM) and reduce their T scores. Consequently, the Softmax Sharding Engine heavily penalized their selection probabilities, successfully suppressing the malicious participation rate down to just 26%. This empirically proves that our system proactively filters our Byzantine threats and minimizes malicious voting influence before the spatial aggregation phase even begins.

Defeating Stealth Temporal Threats. While filtering static Byzantine behavior is crucial, advanced adversaries may attempt to bypass detection using temporal stealth strategies. Because DecentLLMs are "stateless" (treating every consensus round as an isolated event) they are entirely blind to long-term behavioral patterns [1]. To evaluate our system's trustworthiness against sophisticated temporal threats, we subjected the framework to the Inverse Attack and Oscillation Attack. The trajectory of the attacker's Reputation Scores (T) across 50 rounds is plotted in Figure 7 & Figure 8.

Under the Inverse Attack, where adversaries systemati-

cally invert their quality evaluations to perfectly oppose the honest consensus, the EMA update rule instantly identified the maximum spatial divergence. As shown in the Figure 7, this resulted in steep, immediate collapse of the attacker's T scores to 0.150 in the initial rounds, eventually driving them down to a heavily suppressed near 0. This permanently strips their voting weight and effectively neutralized the attack immediately.

More notably, under the Oscillation Attack, adversaries attempt to exploit the system by acting honestly to farm high reputation scores, only to suddenly inject malicious vectors once trusted. The trajectory graph in Figure 8 reveals a violent "bouncing" behavior as the attackers start being malicious. The raw data demonstrates that while attackers successfully farm their reputation to near perfect scores, the EMA mathematical memory (λ) instantly reacts to their subsequent betrayal, reduce their scores down. Because the framework permanently remembers past reputation, oscillating attackers are forced into extended, costly recovery phases, preventing them from ever maintaining enough consistent voting influence to hijack the consensus.

Ultimately, these temporal attack progressions prove that the proposed architecture successfully transforms the decentralized oracle from a stateless, exploitable network into a highly trustworthy, memory-driven meritocracy.

VI. DISCUSSION

A. Broader Architectural Implications

The proposed framework introduces a fundamental paradigm shift in the design of Byzantine-robust leaderless multi-agent systems, successfully transitioning decentralized oracle networks from a reliance on raw egalitarian headcount to a dynamic, meritocratic architecture.

Traditionally, leaderless coordination models such as DecentLLMs operate on an egalitarian premise, dictating that their Byzantine Fault Tolerance (BFT) is strictly bound by the raw count of participating nodes. Under this conventional model, survival mandates that the absolute number of Byzantine attacker remains strictly below the threshold of $f \leq \lfloor \frac{n-1}{2} \rfloor$. If adversaries scale their presence, the system's

safety margin passively decays, leaving the oracle entirely vulnerable to coordinated spatial hijacking once the threshold is crossed. By transforming the standard voting model into a meritocratic system, our architecture permanently breaks this reliance on raw attacker volume. Through the integration of the Reputation-Weighted Geometric Median (WGM) and the Exponential Moving Average (EMA) update rule, the system continuously redistributes spatial influence based on historical reliability. This ensures that the network’s security is governed by an actively defended Honest Weight Fraction (Λ). By aggressively suppressing the influence of malicious vectors, the framework mathematically generates a Meritocratic Safety Margin, enabling the system to proactively survive attacks that would otherwise shatter traditional egalitarian limits.

Furthermore, this meritocratic shift presents a definitive solution to the evaluator reliability problem that currently bottlenecks state-of-the-art leaderless protocol. Because LLM inherently exhibit stochastic variance and occasionally hallucinate, treating all non-Byzantine evaluators equally inherently permits unreliable, high-variance scores to degrade the spatial consensus. Our empirical data proves that the integration of the EMA tracking rule effectively neutralizes this natural variance. By mathematically scaling each agent’s voting influence in proportion to their historical accuracy, the system seamlessly filters out anomalous evaluations. This proves that a decentralized LLM oracle can successfully extract a high accurate and stable consensus signal from an inherently noisy network without sacrificing decentralization.

Ultimately, this architectural shift establishes that future decentralized oracles do not have to compromise between Byzantine robustness, scalability, and output quality, paving the way for highly trustworthy data feeds for smart contract execution.

B. Limitations and Architectural Trade-Offs

While the proposed framework successfully hardens the leaderless oracle against coordinated Byzantine threats and resolves the $O(N^2)$ communication bottleneck, these architectural advancements introduce specific mathematical and physical trade-offs.

The Accuracy vs. Scalability Trade-Off. A primary architectural trade-off of this framework is the slight reduction in peak raw accuracy under uncoordinated noise. As observed in the Random Attack experiments, the baseline achieved marginally higher accuracy peaks (up to 94%) compared to our proposed system (up to 90%). This discrepancy is the direct mathematical cost of the Softmax Sharding Engine. Because the baseline forces all N agents to evaluate every answer, it brute-forces a highly accurate spatial median simply through overwhelming sample size at an unsustainable $O(N^2)$ communication cost. By restricting evaluations to a K -sized committee, our framework intentionally introduces a minor degree of natural variance. However, in distributed systems, trading a nominal 4% accuracy reduction to guarantee $O(K^2)$ scalability and prevent the total structural collapse of the

safety margin under targeted attacks is a highly favorable and necessary exchange.

The "Cold Start" Vulnerability. Furthermore, because the Softmax Sharding and Weighted Geometric Median (WGM) rely entirely on historical reputation, the system is temporarily vulnerable during its initialization phase. When agents first join the network, they are assigned a default reputation score of $T = 0.5$. Consequently, the framework requires a brief "burn-in" period to safely establish its Meritocratic Safety Margin.

Softmax Temperature (τ) Tuning. The Softmax-based probabilistic selection introduces a delicate hyperparameter trade-off via the temperature variable (τ). If τ is set too low, the selection process heavily favors only the highest-reputation experts, maximizing Byzantine robustness but creating an oligarchy that prevents new agents from participating. Conversely, a high τ promotes fairness and network exploration but risks drafting unproven or malicious agents into the evaluation committee.

Absolute Initial Byzantine Weight. While the Reputation-Weighted Consensus permanently breaks the reliance on raw BFT headcounts over time, it is still bound by the initial Honest Weight Fraction threshold ($\Lambda > 0.5$). If a highly coordinated adversary manages to execute an overwhelming Sybil attack (controlling a massive majority of the network’s total reputation weight before the EMA has the opportunity to adjust their scores downward) the initial Byzantine spatial influence could theoretically hijack the WGM instantly.

Prototype and API Constraints. Beyond theoretical limitations, our experimental prototype also faced physical constraints. In a theoretical, production-ready deployment of this decentralized oracle, the multi-agent system would operate on a true peer-to-peer network where each agents is hosted on physically isolated hardware with dedicated GPU resources. This would allow the parallel generation and evaluation phases to execute with true horizontal scalability. However, for the scope of this experiment, the network was simulated locally and routed through a single centralized commercial API (OpenRouter). To prevent the centralized provider from triggering HTTP 429 rate limits, an `asyncio.Semaphore` with a capacity of 30 was implemented to physically throttle the in-flight API requests. Consequently, the absolute execution time recorded in our experiments include artificial queuing latency that would not exist in a true hardware-level deployment.

C. Future Directions

While the current prototype successfully validates the mathematical and architectural foundations of the Reputation-Weighted Consensus Framework, transitioning this system into a production-ready decentralized oracle opens several promising avenues for future systems research.

Cryptoeconomic Security and Staking. Currently, the framework utilizes the internal Reputation Score (T) purely as a mathematical weight for Softmax sharding and spatial aggregation. A critical future direction involves coupling this mathematical reputation directly to a financial incentive layer via blockchain smart contracts. By requiring agents to "stake"

native cryptocurrency to participate in the network, the system could enforce strict economic penalties for Byzantine behavior. If the Exponential Moving Average (EMA) update rule detects malicious spatial divergence and crushes an agent’s T score, the smart contract could be programmed to automatically “slash” their financial stake. This integration would provide an absolute cryptoeconomic security layer, financially ruining attackers on top of mathematically banishing them from the consensus.

Live Mainnet Deployment and True P2P Networking.

As noted in the architectural trade-offs, the current prototype relies on simulated routing through a centralized commercial API to manage the agent pool. Future work will focus on a full-scale deployment of the multi-agent system onto a live, public blockchain mainnet or Layer-2 network. In this true peer-to-peer (P2P) environment, each LLM agent would be hosted on physically isolated hardware with dedicated, decentralized GPU compute. This deployment will allow for the empirical testing of the Softmax Sharding Engine’s true horizontal scalability, physically proving the $O(K^2)$ communication efficiency in a geographically distributed, asynchronous network.

Integrating Multi-Round Debates. Finally, drawing inspiration from broader multi-agent literature, future iterations of the framework could incorporate advanced cognitive coordination techniques such as Multi-Agent Debate (MAD). As demonstrated by recent architectures like the LLM-net framework [11], implementing collaborative and repetitive prompting strategies (where multiple LLM instances propose, critique, and refine solutions through structured debate) can significantly improve mathematical and strategic reasoning, enhance factual validity, and reduce fallacious answers and hallucinations.

VII. CONCLUSION

This research addressed the critical vulnerabilities limiting the deployment of state-of-the-art leaderless multi-agent protocols. While existing leaderless architectures successfully eliminate the single points of failure found in leader-driven models, they introduce a severe “evaluator reliability problem” and a crippling $O(N^2)$ communication bottleneck. To overcome these limitations, we proposed a novel Reputation-Weighted Consensus framework that transforms the standard egalitarian voting model into a scalable meritocracy. By integrating a Softmax Sharding Engine to probabilistically select evaluator committees and an Exponential Moving Average (EMA) update rule to govern a Weighted Geometric Median (WGM), our system mathematically scales an agent’s spatial voting influence strictly based on their historical trustworthiness.

Through rigorous empirical evaluation, we proved that this architecture achieves both superior efficiency and absolute Byzantine fault tolerance. First, the Softmax Sharding Engine successfully flattened the communication bottleneck to $O(K^2)$, significantly reducing mean execution latency by 21% while simultaneously improving baseline consensus accuracy by +2.0 percentage points. Second, during severe 40% network

infiltrations, the sharding engine physically banished attackers, suppressing their committee participation rate from the baseline’s 40% down to just 26.6%. Under extreme 50% Amplification Attacks where the stateless baseline completely collapsed, our framework dynamically constructed a Meritocratic Safety Margin, proactively absorbing the threat and defending spatial center. Finally, we demonstrated that the mathematical memory of the EMA engine permanently neutralized advanced temporal stealth strategies, instantly crushing the reputation trajectories of both Inverse and Oscillation attackers.

Ultimately, by successfully hardening the network against both unintentional variance and coordinated Byzantine attacks, this framework transforms decentralized LLM oracles from exploitable, stateless networks into robust, memory-driven meritocracies.

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APPENDIX

To ensure diversity in both natural language generation and evaluation perspectives, the experimental prototype utilized a heterogeneous pool of 30 LLM sourced via the OpenRouter API

from 9 distinct provider. The models were strictly partitioned into two non-overlapping functional groups.

TABLE II
WORKER AGENT MODEL POOL.

#	Model	Provider
1	openai/gpt-5-nano	OpenAI
2	openai/gpt-oss-20b	OpenAI
3	google/gemini-2.5-flash-lite	Google
4	google/gemma-3-27b-it	Google
5	google/gemma-4-26b-a4b-it	Google
6	nvidia/nemotron-3-super-120b	NVIDIA
7	qwen/qwen3.5-9b	Qwen
8	qwen/qwen3.5-flash-02-23	Qwen
9	qwen/qwen3-coder-next	Qwen
10	minimax/minimax-m2.5	MiniMax
11	z-ai/glm-4.7-flash	Z-AI
12	bytedance-seed/seed-1.6-flash	ByteDance
13	x-ai/grok-3-mini	xAI
14	qwen/qwen3.5-35b-a3b	Qwen
15	upstage/solar-pro-3	Upstage

TABLE III
EVALUATOR AGENT MODEL POOL.

#	Model	Provider
1	deepseek/deepseek-v3.2	DeepSeek
2	deepseek/deepseek-chat-v3.1	DeepSeek
3	x-ai/grok-4.1-fast	xAI
4	x-ai/grok-4-fast	xAI
5	openai/gpt-4.1-nano	OpenAI
6	openai/gpt-4o-mini	OpenAI
7	openai/gpt-oss-120b	OpenAI
8	mistralai/mistral-small-2603	Mistral
9	qwen/qwen3-235b-a22b-2507	Qwen
10	meta-llama/llama-4-maverick	Meta
11	inception/mercury-2	Inception
12	google/gemma-4-31b-it	Google
13	bytedance-seed/seed-2.0-mini	ByteDance
14	xiaomi/mimo-v2-flash	Xiaomi
15	meta-llama/llama-3.3-70b-instruct	Meta